THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(d)** 62°	angles sum to 180°
2	**(d)** obtuse	definition
3	**(c)** 68°	linear pair sums to 180°
4	**(c)** 5	3 × 5 = 15
5	**(a)** 6174	repeated rearrangement process
6	**(a)** 21	each term is the sum of the two before it
7	**(b)** Yes	sum of any two sides must exceed the third
8	**(a)** 21 cm	3 × 7
9	**(d)** 55°	base angles are equal
10	**(a)** 2/5 m²	2/3 × 3/5
11	**(c)** 6/7	multiply by the reciprocal of the divisor
12	**(a)** 3/2	multiply by the reciprocal of the divisor

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	70°, 110°, 110° (2).	2
14	Even, as it ends in 6 (2).	2
15	A number that reads the same both ways (1); 121 and 4554 (1).	1+1
16	60° (2).	2
17	6/5 and 5/6: not equal (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	5x + 20 = 180 (1.5); x = 32 (1.5).	1½+1½
19	$10^2 = 100$ (1.5); pattern n^2 (1.5).	1½+1½
20	x + x + 20 = 90 (1.5); 35° and 55° (1.5).	1½+1½
21	1/3 (3).	3
22	12 pieces (2); nothing remains (1).	2+1
	OR: 1/6 (1.5); 6 (1.5).	1½+1½
23	False (1); acute + acute < 180° (2).	1+2
	OR: 90 − 70 = 20° (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) Last digit even; digit sum by 3; last digit 0/5; digit sum by 9; last digit 0 (2). (ii) Divisible by 2, 3, 5, 9, 10 (1.5). (iii) By 3, 5, 9 (1.5).	2+1½+1½
	OR: (i) (2a + 1) + (2b + 1) = 2(a + b + 1) (1.5). (ii) (2a + 1)(2b + 1) = 2(…) + 1 (1.5). (iii) 63 + 11 = even (2).	1½+1½+2
25	(i) 18 (1). (ii) ₹216 (1). (iii) 14 pieces (1). (iv) 10/3 m (2).	1+1+1+2
	OR: (i) 3/5 (1). (ii) 5/3 × 3/5 = 1 (2). (iii) 25/9 (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** parallel lines	constant distance apart
2	**(a)** right angle	perpendicular
3	**(d)** 17°	angles sum to 90°
4	**(a)** rhythm patterns (long and short beats)	poetic metres
5	**(a)** odd	odd + even
6	**(a)** 20	2n
7	**(a)** one	sum is 180°
8	**(c)** No	sum of any two sides must exceed the third
9	**(a)** 78°	angle sum is 180°
10	**(d)** 1/4	multiply numerators and denominators, then simplify
11	**(d)** smaller than the number	e.g. 1/2 × 8 = 4
12	**(d)** 1	(a/b) × (b/a)

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Parallel lines never meet however far extended (1); intersecting lines meet at a point (1).	1+1
14	The sum is even (1); 7 + 9 = 16 (1).	1+1
15	The three interior angles of a triangle add to 180° (2).	2
16	Altitude is perpendicular from a vertex to the opposite side (1); median joins vertex to midpoint of opposite side (1).	1+1
17	5/6 × 9/10 = 3/4 (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	The lines are parallel (1.5); converse of alternate angles property (1.5).	1½+1½
19	5432 − 2345 = 3087 (3).	3
20	6x = 180, x = 30 (1.5); 40°, 60°, 80° (1.5).	1½+1½
21	5/8 vs 10/9; division (3).	3
22	3x + 10 = 5x − 30 (1.5); x = 20 (1.5).	1½+1½
	OR: Four angles 65° and four angles 115° (3).	3
23	5 + x + 3 divisible by 9: x = 1 (3).	3
	OR: 77 (1.5); palindrome (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) 68° (1). (ii) 68° (1). (iii) 112° (1). (iv) Yes, equal corresponding angles (2).	1+1+1+2
	OR: (i) A line cutting two lines at distinct points (1). (ii) Corresponding, alternate, co-interior, vertically opposite (2). (iii) four of 75°, four of 105° (2).	1+2+2
25	(i) Exterior + interior adjacent = 180 (2). (ii) 90° (1.5). (iii) 90° (1.5).	2+1½+1½
	OR: (i) Base 6; arcs 4 and 5 (2). (ii) Set square (2). (iii) Intersection of altitudes (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** equal	formed by two intersecting lines
2	**(a)** 52°	angles sum to 90°
3	**(d)** 55°	vertically opposite angles are equal
4	**(a)** odd	a product is odd only if both numbers are odd
5	**(b)** even	odd + odd
6	**(c)** 4	6 × 4 = 24
7	**(a)** 70°	angle sum is 180°
8	**(b)** orthocentre	concurrent
9	**(c)** Yes	sum of any two sides must exceed the third
10	**(a)** 3/2	multiply by the reciprocal of the divisor
11	**(c)** greater than the number	e.g. 8 ÷ 1/2 = 16
12	**(b)** 45	‘of’ means multiply

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	If corresponding (or alternate) angles are equal (2).	2
14	3 + 6 + 5 + 4 = 18 (1); yes (1).	1+1
15	Acute-angled, right-angled, obtuse-angled (2).	2
16	24 (2).	2
17	Two halves fit in each whole (2).	2

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	9x = 180, x = 20 (1.5); 80°, 100°, 80°, 100° (1.5).	1½+1½
19	e.g. 25 + 25 = 50 (A=2, B=5, C=0) (3).	3
20	(36 − 10) ÷ 2 = 13 cm (3).	3
21	3/4 × 5/2 = 15/8 = 1 7/8 cups (3).	3
22	Even; it has the even factors 14 (3).	3
	OR: 9, 18, 27 … digits add to 9 (3).	3
23	Each vertex joins the midpoint of the opposite side (1.5); gives three medians (1.5).	1½+1½
	OR: Place a set square on a side (1); slide so it touches the opposite vertex (1); draw perpendicular (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) 1, 2, 3, 5, 8, 13, 21, 34, 55, 89 (2). (ii) Add previous two (1). (iii) 11th term (2).	2+1+2
	OR: (i) 15 (1). (ii) Even numbers 2, 4, 6, 8 (1). (iii) 2 7 6 / 9 5 1 / 4 3 8 (2). (iv) 2 + 5 + 8 = 15 (1).	1+1+2+1
25	(i) 7 1/2 (2). (ii) 9 (2). (iii) as shown (1).	2+2+1
	OR: (i) 6 km/h (2). (ii) 15 km (2). (iii) 4 1/2 h (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.